

BANK ACCOUNT SYSTEM

Objective / Aim

To design and implement a Bank Account System in Java demonstrating object-oriented programming concepts such as attributes, methods, constructors, object creation, inheritance, method overriding, deposits, withdrawals, interest calculation, and fund transfers.

Problem Statement

Design a bank account system using a base Account class and derived SavingsAccount and CurrentAccount classes. The system should support deposits, withdrawals, interest calculation, overdraft facility, fund transfers, and display of account details before and after transactions.

Algorithm / Logic

1. Create the Account base class.
2. Declare account number, account holder name, balance, and account type.
3. Create a constructor to initialize the account details.
4. Define deposit(), withdraw(), transfer(), and displayAccountDetails().
5. Create SavingsAccount that inherits Account and add interestRate.
6. Define calculateInterest() in SavingsAccount.
7. Override displayAccountDetails() in SavingsAccount to display the interest rate.
8. Create CurrentAccount that inherits Account and add overdraftLimit.
9. Override withdraw() in CurrentAccount to allow withdrawal up to the overdraft limit.
10. Override displayAccountDetails() in CurrentAccount to display the overdraft limit.
11. Create SavingsAccount and CurrentAccount objects.
12. Display the initial account details.
13. Perform deposit and withdrawal operations.
14. Calculate interest for the savings account.
15. Transfer funds from the savings account to the current account.
16. Display the final account details and final balances.
17. Stop the program.

Steps for Execution

1. Create a Java project and create a file named BankAccountDemo.java.
2. Copy the given source code into the file.
3. Compile and run the program.
4. The program creates SavingsAccount and CurrentAccount objects.
5. Observe the initial account details.
6. Perform the deposit operations.
7. Perform the withdrawal operations, including the current-account overdraft facility.
8. Calculate the savings-account interest.
9. Transfer Rs. 2000 from the savings account to the current account.

10. Observe the final account details and final balances.

Source Code

```
// Base class
class Account {
    // Attributes
    protected int accountNumber;
    protected String accountHolderName;
    protected double balance;
    protected String accountType;

    // Constructor
    Account(int accountNumber, String accountHolderName,
            double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.accountHolderName = accountHolderName;
        this.balance = balance;
        this.accountType = accountType;
    }

    // Deposit method
    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Deposited: Rs. " + amount);
        } else {
            System.out.println("Invalid deposit amount.");
        }
    }

    // Withdraw method
    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance -= amount;
            System.out.println("Withdrawn: Rs. " + amount);
        } else {
            System.out.println("Insufficient balance or invalid amount.");
        }
    }

    // Transfer method
    public void transfer(Account receiver, double amount) {
        if (amount > 0 && amount <= balance) {
            this.balance -= amount;
            receiver.balance += amount;

            System.out.println("Transferred: Rs. " + amount);
            System.out.println("From Account: " + this.accountNumber);
            System.out.println("To Account: " + receiver.accountNumber);
        } else {
            System.out.println("Transfer failed: Insufficient balance.");
        }
    }

    // Display account details
    public void displayAccountDetails() {
        System.out.println("Account Number      : " + accountNumber);
        System.out.println("Account Holder   : " + accountHolderName);
        System.out.println("Account Type     : " + accountType);
        System.out.println("Balance          : Rs. " + balance);
    }
}

// SavingsAccount inherits Account
class SavingsAccount extends Account {
    private double interestRate;

    SavingsAccount(int accountNumber, String accountHolderName,
            double balance, double interestRate) {
        super(accountNumber, accountHolderName, balance, "Savings");
        this.interestRate = interestRate;
    }

    public void calculateInterest() {
        double interest = balance * interestRate / 100;
        balance += interest;

        System.out.println("Interest Rate      : " + interestRate + "%");
        System.out.println("Interest Earned    : Rs. " + interest);
    }
}

@Override
```

```

    public void displayAccountDetails() {
        super.displayAccountDetails();
        System.out.println("Interest Rate      : " + interestRate + "%");
    }
}

// CurrentAccount inherits Account
class CurrentAccount extends Account {
    private double overdraftLimit;

    CurrentAccount(int accountNumber, String accountHolderName,
                    double balance, double overdraftLimit) {
        super(accountNumber, accountHolderName, balance, "Current");
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    public void withdraw(double amount) {
        if (amount > 0 && amount <= balance + overdraftLimit) {
            balance -= amount;

            System.out.println("Withdrawn: Rs. " + amount);
            System.out.println("Withdrawal allowed using overdraft facility.");
        } else {
            System.out.println("Withdrawal failed: Overdraft limit exceeded.");
        }
    }

    @Override
    public void displayAccountDetails() {
        super.displayAccountDetails();
        System.out.println("Overdraft Limit    : Rs. " + overdraftLimit);
    }
}

// Main class
public class BankAccountDemo {
    public static void main(String[] args) {

        SavingsAccount savings = new SavingsAccount(
            101, "Rahul", 10000, 5.0);

        CurrentAccount current = new CurrentAccount(
            102, "Arun", 5000, 3000);

        System.out.println("===== INITIAL ACCOUNT DETAILS =====");

        System.out.println("\nSavings Account:");
        savings.displayAccountDetails();

        System.out.println("\nCurrent Account:");
        current.displayAccountDetails();

        System.out.println("\n===== DEPOSIT =====");

        savings.deposit(2000);
        current.deposit(3000);

        System.out.println("\n===== WITHDRAWAL =====");

        savings.withdraw(1500);
        current.withdraw(9000);

        System.out.println("\n===== INTEREST CALCULATION =====");

        savings.calculateInterest();

        System.out.println("\n===== FUND TRANSFER =====");

        savings.transfer(current, 2000);

        System.out.println("\n===== FINAL ACCOUNT DETAILS =====");

        System.out.println("\nSavings Account:");
        savings.displayAccountDetails();

        System.out.println("\nCurrent Account:");
        current.displayAccountDetails();

        System.out.println("\n===== FINAL BALANCES =====");

        System.out.println("Savings Account Balance : Rs. " + savings.balance);
        System.out.println("Current Account Balance : Rs. " + current.balance);
    }
}

```

} }